

AI-Powered Credit Risk Intelligence Platform

Explainable Credit Risk Decision Support with ML, EDA & Natural-Language Analytics

Tejas R M

MSc Artificial Intelligence & Machine Learning

Class Imbalance · TreeSHAP Explainability · Conversational Analytics

Use Case

Platform Objective
"Design and build a lightweight AI-powered credit risk platform that combines data analysis, machine learning, explainability and conversational analytics."

Use Case Scope

- **Credit Risk Assessment:**
Using the Home Credit Default Risk dataset (307k+ loan applicants).
- **Pattern & Insight Discovery:**
Identify relationships from applicant, credit, and repayment history.
- **Predictive Default Modeling:**
Train and evaluate an ML model to predict credit default risk.
- **Explainable Predictions:**
Provide transparent, applicant-level feature attributions.
- **Natural-Language Analytics:**
Enable conversational portfolio exploration via Talk-to-Data.

AI Engineering Modules

- 1. Data Understanding & EDA**
Data quality, distributions & 5+ key business insights
- 2. Machine Learning Layer**
Class imbalance handling, risk score & tier classification
- 3. Explainable AI (XAI)**
TreeSHAP local feature attribution for decision transparency
- 4. Talk-to-Data System**
NL-to-SQL query engine with safety validation & plain-English answers
- 5. User Interface & Docker**
Lightweight multi-section web UI & one-command containerization

Dataset & Key EDA Findings

307,511

TOTAL APPLICANTS

24,825

DEFAULTS (TARGET = 1)

8.07%

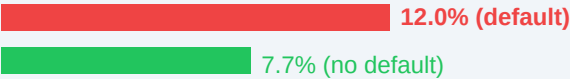
BASELINE DEFAULT RATE

Key EDA Insights — Observed Default Rates

Previous Refusals



High POS/CASH DPD



High CC Utilization



Frequent Late Payments

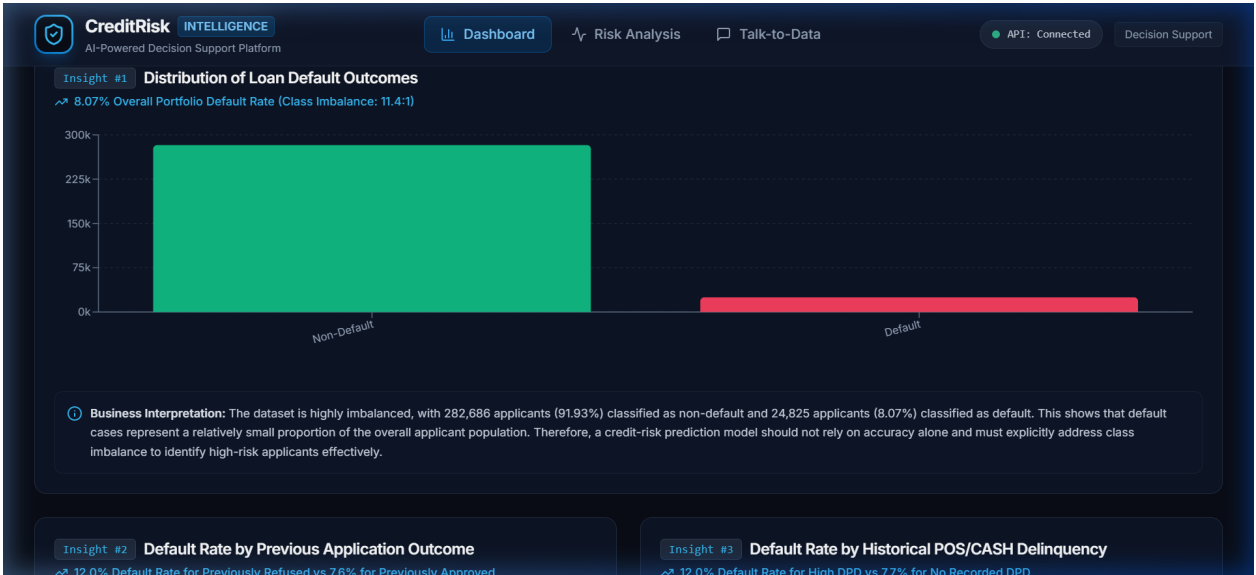


Low Payment Ratio



*Observed historical associations — not causal.

Portfolio EDA Dashboard



Machine Learning Approach & Model Selection



Model Comparison (Validation Set N = 61,503)

Model	ROC-AUC	PR-AUC
Logistic Regression	0.7735	0.2608
XGBoost	0.7874	0.2848
LightGBM	0.7870	0.2844
CatBoost ✓ (Selected)	0.7878	0.2862

PR-AUC is the primary metric — chosen for severe class imbalance (8.07% defaults).

Why CatBoost?

Highest PR-AUC:
0.2862 — best at identifying minority defaults.

Highest ROC-AUC:
0.7878 — best overall ranking discrimination.

Native Categorical:
Handles high-cardinality variables without one-hot explosion.

Operating Threshold:
Tuned to 0.66 on validation set to maximise F1 score.

Class Distribution (Validation Set)
Non-default: 91.93% | Default: 8.07%

Final Model Results & Risk Scoring

0.7878

ROC-AUC

0.2862

PR-AUC

0.2824

Precision

0.4354

Recall

0.3426

F1 Score

0.8651

Accuracy

Confusion Matrix (N = 61,503)

	Pred 0 (Non-default)	Pred 1 (Default)
Actual 0	51,044 (TN)	5,494 (FP)
Actual 1	2,803 (FN)	2,162 (TP) ✓

Operating Threshold = 0.66

Tuned on validation data to maximise F1.

True Positive Rate: 43.5%

Of actual defaults correctly flagged

False Positive Rate: 9.7%

Non-defaults incorrectly flagged

Precision: 28.2%

Flagged applicants that truly default

F1 Score: 0.343

Harmonic mean of precision & recall

PR-AUC is the primary metric — chosen for severe class imbalance.

Risk Score = Probability x 100

LOW

< 30

Approve / Low Risk

MEDIUM

30–<60

Manual Review Required

HIGH

>= 60

High Risk / Reject Recommendation

Verified Benchmark Applicants

ID 396899:

0.228 -> 22.83

LOW

Approve

ID 100002:

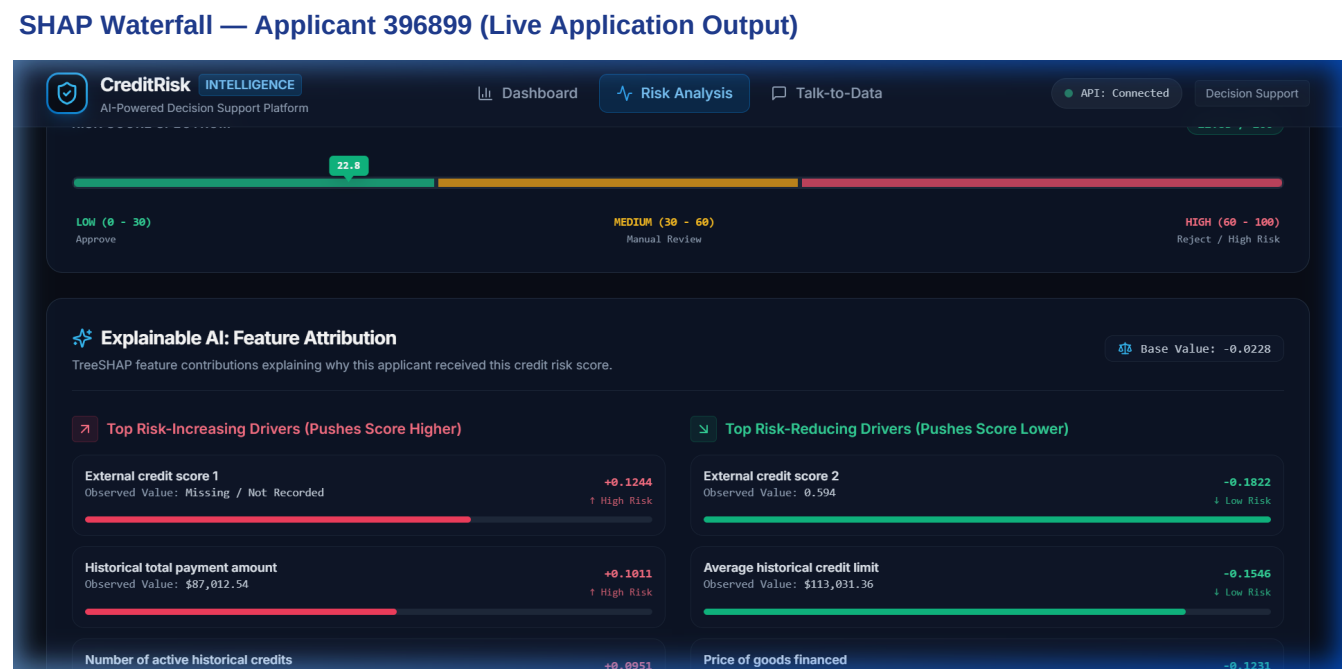
0.828 -> 82.76

HIGH

Reject Recommendation

*Decision-support — not autonomous lending.

Explainable AI with SHAP



- **Base Value (Expected log-odds):**
-0.0228 (corresponds to ~49.4% base rate logit).
- **Model Output (Applicant 396899):**
Default probability 0.2282 (Risk Score: 22.83 / Low Risk).
- **Top Positive Driver (+Risk):**
TOTAL_PAYMENT_AMOUNT (+0.1244) & AMT_ANNUITY (+0.1011).
- **Top Negative Driver (-Risk):**
EXT_SOURCE_2 (-0.1822) & BUREAU_AVG_LIMIT (-0.1546).

TreeSHAP Explainability

- **Local Attribution:**
Exact applicant-level contributions computed per prediction.
- **Positive value:**
Feature increases predicted default risk.
- **Negative value:**
Feature reduces predicted default risk.
- **Fair-Lending Guardrail:**
Protected demographic attributes are excluded from user-facing individual explanations.

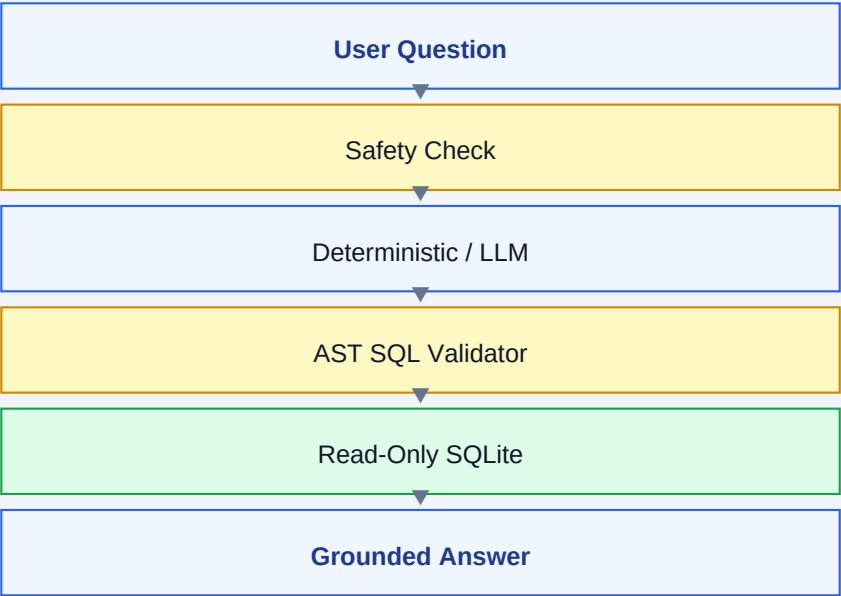
Applicant 396899 — Key Drivers

- ↑ **Risk-Increasing**
 - TOTAL_PAYMENT_AMOUNT
 - BUREAU_ACTIVE_COUNT
 - AMT_ANNUITY
- ↓ **Risk-Reducing**
 - EXT_SOURCE_2
 - BUREAU_AVG_LIMIT
 - AMT_GOODS_PRICE

*SHAP = model contribution, not real-world causation.

Talk-to-Data: Natural-Language Analytics

NL → SQL Flow



Example Questions

- "What is the default rate?"
- "How many applicants have income > 300,000?"
- "Show top 5 high-risk applicants."

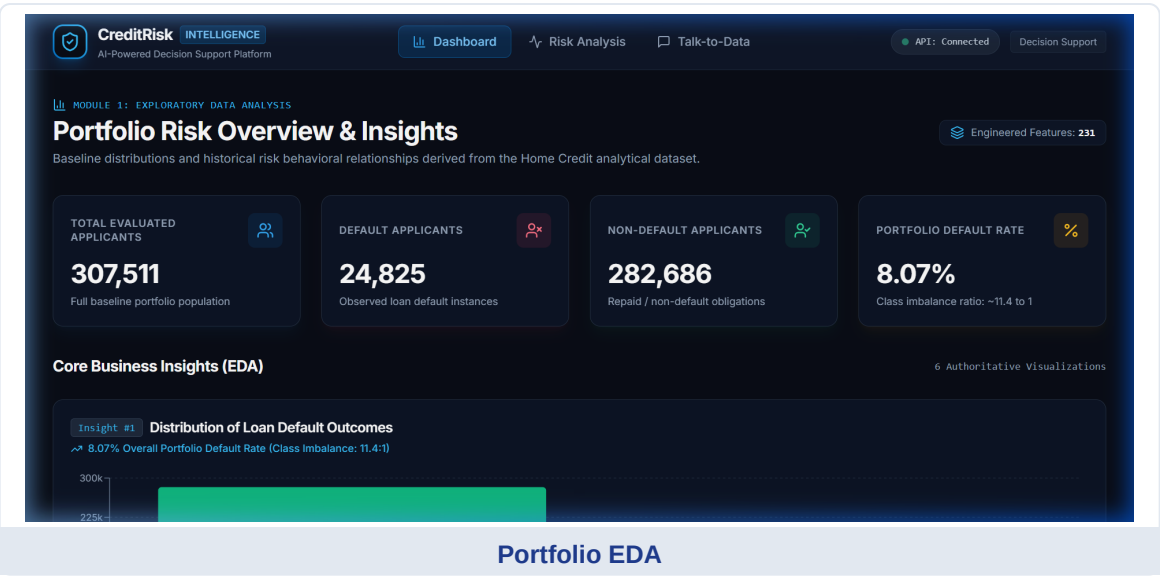
Live Talk-to-Data Query — Top 5 High-Risk Applicants

The screenshot shows the 'CreditRisk INTELLIGENCE' dashboard. The 'Talk-to-Data' tab is active. A welcome message states: "Welcome to Talk-to-Data! I can analyze the 307,511 applicant portfolio, generate schema-grounded SQL, and synthesize executive business insights. How can I assist with your portfolio analysis today?" Below this, a button says "Show me the top 5 high-risk applicants". The response shows a summary: "Here are the 5 applicants with the highest CatBoost model-predicted default risk. The highest-risk applicant (ID: 181065) has a predicted default probability of 98.7% (Risk Score: 98.7014, Band: High)." Below the summary, a table displays the top 5 high-risk applicants.

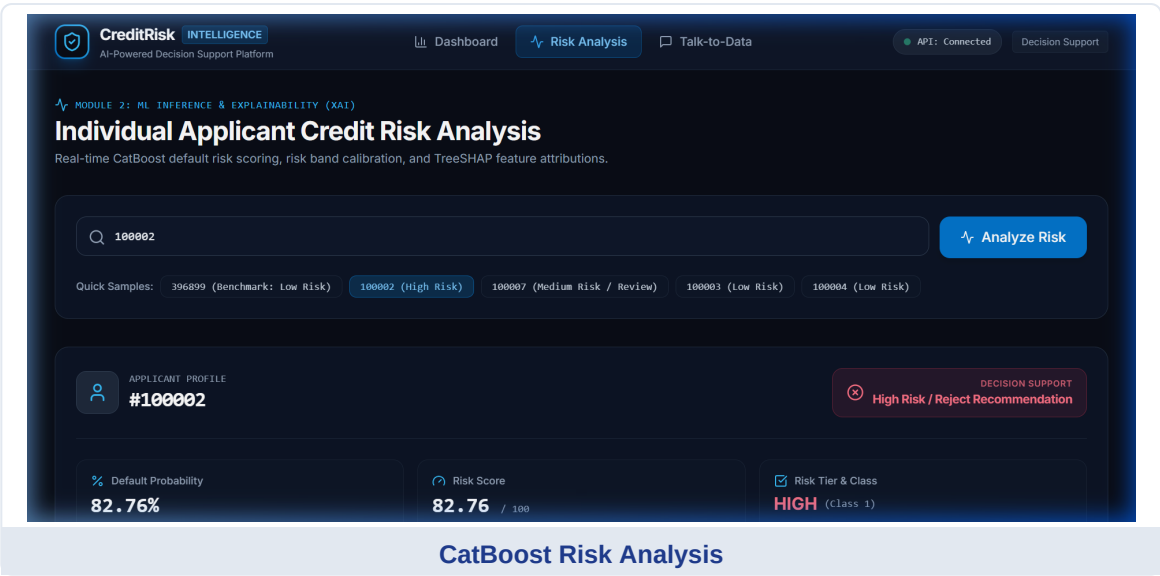
SK_ID_CURR	RISK_SCORE	DEFAULT_PROBABILITY	RISK_BAND	AMT_INCOME_TOTAL	AMT_CREDIT	NAME_EDUCATION_TYPE	NAME_INCOME_TYPE
181,065	98.70	0.99	High	67,500	436,032	Secondary / secondary special	Working
304,421	98.46	0.98	High	126,000	364,896	Incomplete higher	Working

- **Deterministic fast-path:**
Intent matching without LLM for speed & reliability.
- **Schema-aware LLM path:**
LLM generates SQL aware of actual table schema.
- **AST safety check:**
All SQL validated via abstract syntax tree before execution.
- **Read-only SQLite:**
No write operations possible — analytics DB is protected.

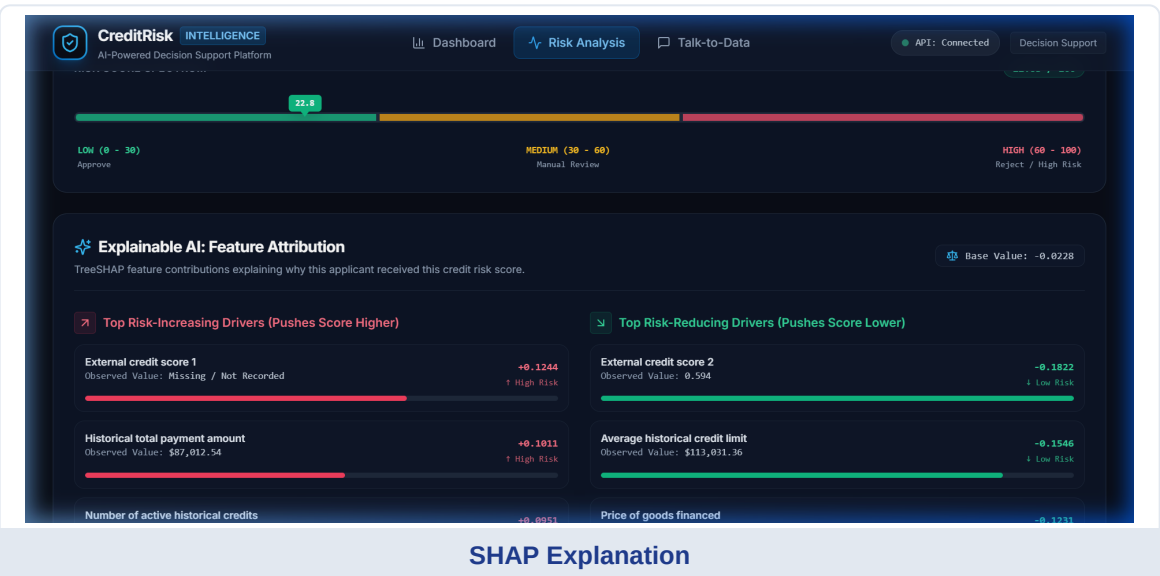
Application Outputs



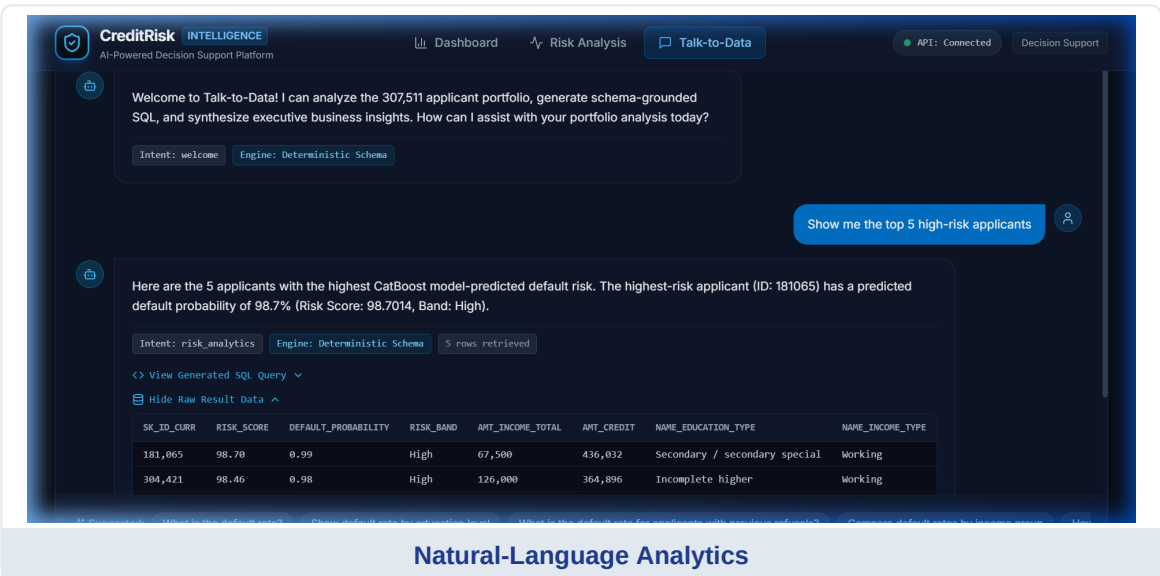
Portfolio EDA



CatBoost Risk Analysis



SHAP Explanation



Natural-Language Analytics

Design Decisions, Limitations & Conclusion

Major Design Decisions

- **CatBoost + PR-AUC:**
Highest PR-AUC (0.2862) on 8.07% imbalanced class.
- **Threshold 0.66:**
Tuned on validation set to maximise F1 score.
- **Dual SQLite Stores:**
Analytical + inference DBs — eliminates 2.64 GB raw CSV at runtime.
- **Pre-scored Risk Table:**
All 307,511 applicants scored for instant portfolio queries.
- **Hybrid NL-to-SQL:**
Deterministic fast path + schema-aware LLM for safe analytics.
- **TreeSHAP:**
Local attribution with demographic masking as a fair-lending guardrail.

Limitations

- **Historical Dataset:**
Static snapshot; model behaviour on new distributions not tested.
- **Single Split:**
One stratified split used; k-fold cross-validation not performed.
- **Moderate PR-AUC:**
0.2862 reflects inherent difficulty of rare-event default prediction.
- **No Calibration:**
Probability calibration (Platt / Isotonic) not yet applied.
- **Fairness Audit:**
Formal statistical fairness tests not yet implemented.
- **LLM Dependency:**
Conversational queries need LLM uptime; deterministic fallback available.

Future Work & Ethics

- **Prob. Calibration:**
Platt scaling or isotonic regression for well-calibrated scores.
- **Fairness Testing:**
Statistical parity, equalised odds across demographic groups.
- **Cross-Validation:**
k-fold evaluation for more robust generalisation estimates.
- **Drift Detection:**
Flag feature distribution shift on incoming applicant batches.
- **Prototype Status:**
Academic decision-support tool — not a production lending system.

"EDA + Explainable ML + Natural-Language Analytics — one credit-risk decision-support platform."

AI-assisted decision support — not autonomous lending.